

Desk Research

Problem Statement:

Credit-card default is a credit-risk problem in which a financial institution needs to assess whether a client is likely to fail to make their required payments. Our project focuses on predicting whether a credit-card client will default on their payment in the following month. We use the UCI “Default of Credit Card Clients” dataset, which contains 30,000 clients and 23 features and represents customers in Taiwan. The dataset was created for research into predicting the probability of customer default from a risk-management perspective.

SDG 8 - Decent Work and Economic Growth:

This problem is relevant to SDG 8 - Decent Work and Economic Growth. SDG 8 includes promoting inclusive economic growth and, specifically under Target 8.10, strengthening financial institutions and expanding access to financial services. Target 8.3 also explicitly connects economic development with access to financial services. Therefore, our project examines how machine-learning-based credit-risk prediction could support financial institutions in managing credit risk while operating within a broader goal of responsible and inclusive access to financial services.

User-group:

The primary user: is a credit-risk analyst at a bank or credit-card provider.

Affected users: credit-card clients.

The prediction would support the analyst in identifying clients who may require further risk assessment or risk-management action. However, the model should not automatically be assumed to apply to all credit-card clients internationally: the dataset represents a specific population in Taiwan, so its representativeness for another population must be evaluated as part of the project.

False Positive vs. False Negative

A false positive occurs when the model predicts that a client will default, but the client actually does not default. This could result in the client being treated as higher risk than they actually are and may lead to unnecessary risk-management actions. A false negative

occurs when the model predicts that a client will not default, but the client actually does default. This can expose the financial institution to financial losses because the client's credit risk was underestimated. For our project, we consider avoiding false negatives particularly important because missing a client who actually defaults can expose the financial institution to credit losses. Therefore, we will use recall for the default class as an important evaluation metric, because recall measures how many of the clients who actually defaulted were correctly identified by the model.

Main Evaluation Metric

The main evaluation metric for our project will be **recall for the default class**. We chose recall because our main priority is to identify as many clients who actually default as possible. A false negative occurs when the model predicts that a client will not default, but the client actually does default. A high recall means that the model correctly identifies a larger proportion of the clients who actually default, reducing the number of missed defaults. We will still report other metrics, such as precision and F1-score, to provide a more complete evaluation of the models, but recall will be our primary metric because we place greater importance on avoiding false negatives.

Dataset Justification

We selected the Default of Credit Card Clients dataset from the UCI Machine Learning Repository. The dataset was created by I-Cheng Yeh and is associated with a 2009 research paper by I. Yeh and Che-hui Lien. The research focuses on predicting customer default payments in Taiwan from a credit-risk management perspective. The dataset contains 30,000 instances and 23 features and is a binary classification problem, where the target represents whether a client defaulted on their payment (1 = yes, 0 = no). The data represents credit-card clients in Taiwan and includes information such as credit limit, demographic characteristics, repayment history, bill amounts, and previous payments. The UCI page reports no missing values. The dataset is licensed under CC BY 4.0, which allows sharing and adaptation provided that appropriate credit is given.

The dataset is suitable for our hackathon because it is a real, documented classification dataset with more than the required number of rows and usable features. It also addresses credit and debt risk, which is directly relevant to economic security and therefore provides a concrete problem that we can connect to SDG 8. However, the dataset represents customers in Taiwan, so we cannot assume that a model trained on this data will

automatically generalize to customers in the Netherlands or other populations. This limitation will be considered during our evaluation and ethical reflection.

Ethical Reflection:

This data set contains financial and demographic information from credit card clients in Taiwan. Even though the UCI page reports no missing values, we still need to consider potential risks and impacts. A model trained on Taiwanese client data might not apply fairly or accurately to people in other countries like the Netherlands. We must be cautious about fairness and bias, especially if demographic features like age, education, or relationship status could lead to unequal treatment. We should avoid using the model to make critical decisions about individuals without safeguards, explainability, and human oversight.

Sources:

UCI Machine Learning Repository - Default of Credit Card Clients: [UCI Machine Learning Repository](#)

United Nations - SDG 8: Decent Work and Economic Growth: sdgs.un.org